Mean income

MATH 113

Group 1

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

2. What parameter can we estimate using this statistic?

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and _______ pieces of card.
2. Write _______ on each piece of card.
3. __________
4. Repeatedly, sample *(in what special way?)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card __________
7. Stop sampling from the cup when I have selected ____ pieces of card.
8. My collection of __________ constitutes my bootstrap sample.
9. Then compute the _________ and plot it on a ______.
10. Do this over and over again _______ times. The resulting distribution is called a ______________.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

6. Where is it centered?

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

9. Complete the sentence of interpretation:

We are ___% confident that the population mean income of _________ lies between ______ and _______.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

Group 2

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why? Because it is based on a sample, not the population.

2. What parameter can we estimate using this statistic? We can estimate the mean income of all US adults in 2012.

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

43804.82

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and 500 pieces of card.
2. Write income value on each piece of card.
3. Shuffle the cards.
4. Repeatedly, sample with replacement from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card back.
7. Stop sampling from the cup when I have selected 500 pieces of card.
8. My collection of cards constitutes my bootstrap sample.
9. Then compute the bootstrap statistic and plot it on a dotplot.
10. Do this over and over again multiple times. The resulting distribution is called a sampling distribution.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

46982.68 (mean income)

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

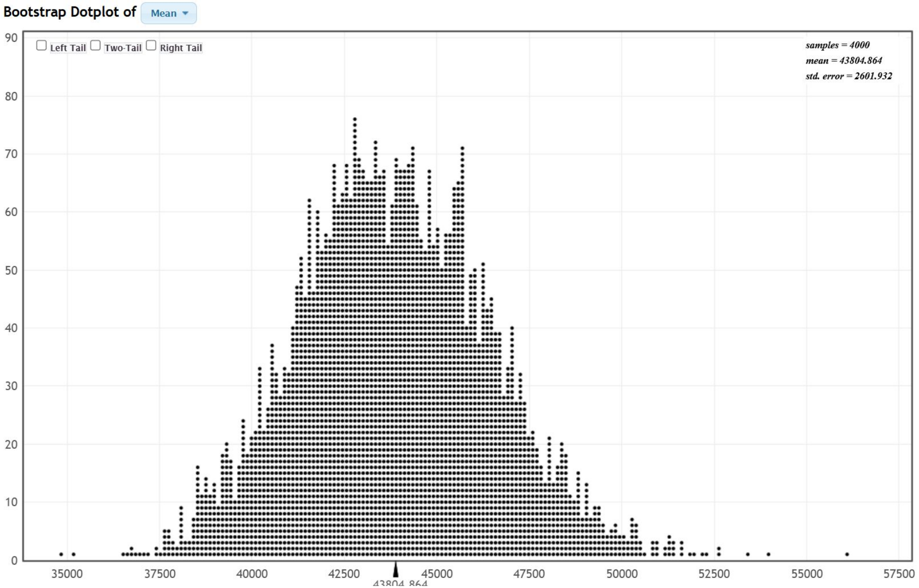
Symmetric, has a bell curve

6. Where is it centered?

43804.864

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

Yes!



7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

5109.52 from center

9. Complete the sentence of interpretation:

We are 95% confident that the population mean income of 43804.864 lies between 38852.37 and 49071.41.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

Group 3

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

   The mean is a statistic because it is calculated from a sample of 500 US adults.

2. What parameter can we estimate using this statistic?

   We can use the mean calculated to estimate the true mean income of all US adults in 2012.

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and ______ pieces of card.
2. Write ______ on each piece of card.
3. ________
4. Repeatedly, sample *(in what special way?)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card ________
7. Stop sampling from the cup when I have selected ____ pieces of card.
8. My collection of ________ constitutes my bootstrap sample.
9. Then compute the ________ and plot it on a _____.
10. Do this over and over again ______ times. The resulting distribution is called a __________.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

6. Where is it centered?

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

9. Complete the sentence of interpretation:

We are ___% confident that the population mean income of _________ lies between ______ and _______.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

Group 4

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?
    - Because this is representative of an attribute attributive of the representative sample, not the population as a whole, and there is no indication at this point that this could be applied to the population
2. What parameter can we estimate using this statistic?
    - We can estimate the population's mean income from that statistic because the sample is a representative of the national population.

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

$43,804.82

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

Get a cup and ____different__ pieces of card.

1. Write __an income_____on each piece of card.
2. ___Shuffle the cards in the cup_______
3. Repeatedly, sample randomly from the cup.
4. After I pull a card out of the cup, record the income value.
5. Then, put the piece of card __back in the cup and reshuffle________
6. Stop sampling from the cup when I have selected _500___pieces of card.
7. My collection of __card pieces________constitutes my bootstrap sample.
8. Then compute the __mean_______ and plot it on a __scatter plot___.
9. Do this over and over again __1000_____ times. The resulting distribution is called a Sampling distribution.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

    $45,606.86

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

    Bell shaped

6. Where is it centered?

    $43,761.72

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

We're within $50 so it's close enough for government work.

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

$38,809.48 and $49,029.71

9. Complete the sentence of interpretation:

We are _95__% confident that the population mean income of ___us adults in 2012______ lies between __$38,809.48 ____ and __$49,029.71_____.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval: $41,949.43 and $45,485.11

11. Write a sentence of interpretation: We are confident that 50% of the 2012 adult population has a mean income that lies between $41,949.43 and $45,485.11.

Group 5

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?
2. What parameter can we estimate using this statistic?

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and ______ pieces of card.
2. Write ______ on each piece of card.
3. __________
4. Repeatedly, sample *(in what special way?)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card __________
7. Stop sampling from the cup when I have selected ____ pieces of card.
8. My collection of __________ constitutes my bootstrap sample.
9. Then compute the ________ and plot it on a _____.
10. Do this over and over again ______ times. The resulting distribution is called a _____________.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

6. Where is it centered?

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

9. Complete the sentence of interpretation:

We are ___% confident that the population mean income of _________ lies between ______ and _______.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

Group 6

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

   This is a statistic because it is taken from a sample, not a population.

2. What parameter can we estimate using this statistic?

The entire adult population of Americans with an income in 2012.

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

$43,804.82

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and 500 pieces of card.
2. Write a salary on each piece of card.
3. _put them in cup, shuffle_
4. Repeatedly, sample *(blindfolded)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card back into the cup
7. Stop sampling from the cup when I have selected 500 pieces of card.
8. My collection of 500 constitutes my bootstrap sample.
9. Then compute the __mean_ and plot it on a dotplot
10. Do this over and over again 1000 times. The resulting distribution is called a _Sampling distribution__.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

$45002.98

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

Normal (symmetric)

6. Where is it centered?

$43811.979

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

Yes

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

$43,811.979

9. Complete the sentence of interpretation:

We are 95% confident that the population mean income of salaries lies between $38928.210 and $49005.030.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval: 42040.920- 45500.440

11. Write a sentence of interpretation:

We are 50% confident that the population mean income of salaries lies between 42040.920- 45500.440.

Group 7

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

   Because it’s a sample statistic, not a population

2. What parameter can we estimate using this statistic?

   x-bar

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?
43623.09

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and _500_ pieces of card.
2. Write _an income_ on each piece of card.
3. __________
4. Repeatedly, sample _randomly_ from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card _back_
7. Stop sampling from the cup when I have selected _500_ pieces of card.
8. My collection of _500_ constitutes my bootstrap sample.
9. Then compute the _mean_ and plot it on a _dotplot.
10. Do this over and over again _1000_ times. The resulting distribution is called a_bootstrap_.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Mean of income

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

Normal bell curve

6. Where is it centered?

43875.84

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

Yes, but there are small peaks spread throughout the sample

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

39057 - 49235

9. Complete the sentence of interpretation:

We are 95% confident that the population mean income of _US adults in 2012_ lies between _$39057_ and _$49235_.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval: 42078 - 45520

11. Write a sentence of interpretation: We are 50% confident that the population mean income of US adults in 2012 lies between $42078 and $45520

Group 8

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?
2. What parameter can we estimate using this statistic?

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and ______ pieces of card.
2. Write ______ on each piece of card.
3. ________
4. Repeatedly, sample *(in what special way?)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card ________
7. Stop sampling from the cup when I have selected ____ pieces of card.
8. My collection of ________ constitutes my bootstrap sample.
9. Then compute the ________ and plot it on a _____.
10. Do this over and over again ______ times. The resulting distribution is called a __________.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

6. Where is it centered?

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

9. Complete the sentence of interpretation:

We are ___% confident that the population mean income of _________ lies between ______ and _______.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

Group 9

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

   This would be a statistic because we are calculating a number, as in the mean income, which would be a statistic about our sample of 500 US adults

2. What parameter can we estimate using this statistic?

A parameter that you can be used for this statistic is that a us adult has to be able to make 40,000

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

The sample mean income for the 500 american adults is 43804.82

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and __500____ pieces of card.
2. Write __income_____on each piece of card.
3. __shuffle_______
4. Repeatedly, sample *(in what special way?)* from the cup. Blindly
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card _back in the cup_________
7. Stop sampling from the cup when I have selected _500___pieces of card.
8. My collection of ___cards_______constitutes my bootstrap sample.
9. Then compute the __bootstrap statistic_______ and plot it on a _scatterplot____.
10. Do this over and over again __10_____ times. The

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

46, 327.86

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

The shape of the distribution seems to be centered in which it is bell shaped

6. Where is it centered?

The center of the 43, 798.80

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

Yes it is exactly what had happened in this case.

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

38,792.39-49,071.96= 10, 279.57

9. Complete the sentence of interpretation:

We are _95__% confident that the population mean income of _43798.80________ lies between _38792.39_____ and __49071.96_____.

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval:

11. Write a sentence of interpretation:

The graph of the mean income of 95% of adults in America is 43.798.80.

Group 10

I have a dataset containing the annual income for a nationally representative sample of 500 US adults in 2012. This dataset is based on the ACS12 dataset provided by our textbook authors.

1. We can use technology to compute the mean income of all individuals in the dataset. This would be considered a statistic as opposed to a parameter. Why?

   Because we are taking the average of a sample

2. What parameter can we estimate using this statistic?

The mean income for all US adults

Download the dataset from Blackboard. Go to Statkey > Bootstrap Confidence Intervals > CI for Single Mean, Median, St. Dev. Click Upload File (one of the top buttons), find the dataset in your Downloads folder and upload it.

We are interested in using this random sample of incomes to find an interval estimate for the population mean income for all US adults.

3. What is the sample mean income?

43804.82

PROBLEM: We don't have access to the population so we can't repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the 95% confidence interval. (Or calculate the 25th and 75th percentiles to compute the 50% confidence interval.)

SOLUTION: We use the technique of BOOTSTRAPPING.

Last time, you might remember we created our first bootstrap sample for the proportion of yellow M&Ms using colored counters and a cup. We won't do this sort of thing this time, but if we did, what are the steps we would have to take?

1. Get a cup and 500 pieces of card.
2. Write incomes on each piece of card.
3. __________
4. Repeatedly, sample *(in what special way?)* from the cup.
5. After I pull a card out of the cup, record the income value.
6. Then, put the piece of card back
7. Stop sampling from the cup when I have selected 500 pieces of card.
8. My collection of incomes constitutes my bootstrap sample.
9. Then compute the mean and plot it on a dotplot.
10. Do this over and over again _______ times. The resulting distribution is called a bootstrap sample distribution.

Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters.

4. What is the bootstrap statistic in this case?

Mean income of all americans

Try this a few more times.

Ask StatKey to generate 4000 more samples. Answer the questions:

5. What shape does the bootstrap distribution have?

Bell shaped

6. Where is it centered?

Centered at 44,000

9. Typically, for a sufficiently large sample size, the bootstrap distribution will be symmetric, bell-shaped and centered at the original sample statistic. Is this what happened in your case? (Check and report below.)

yes

7. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

8. What is our 95% confidence interval for the population mean income of all US adults in 2012?

(38982.452, 49145.993)

9. Complete the sentence of interpretation:

We are **95%** confident that the population mean income of **US Adults** lies between **38982.452** and **49145.993.**

10. Go back to Statkey and change the proportion that appears in the center so that you are computing a 50% confidence interval instead.

Report the confidence interval: 42035.395, 45552.195

11. Write a sentence of interpretation:

We are **50%** confident that the population mean income of US adults lies between **42035.395** and **45552.195**